

(b) Values of s and w are given in Table 2.1.

Table 2.1

s/mm	$\frac{1}{s}/\text{mm}^{-1}$	w/mm	y/mm
0.18 ± 0.01		33.0	
0.21 ± 0.01		28.9	
0.24 ± 0.01		25.1	
0.27 ± 0.01		22.6	
0.31 ± 0.01		19.6	
0.38 ± 0.01		15.9	

Calculate and record values of $\frac{1}{s}/\text{mm}^{-1}$ and y/mm in Table 2.1. Include the absolute uncertainties in $\frac{1}{s}$. [2]

(c) (i) Plot a graph of y/mm against $\frac{1}{s}/\text{mm}^{-1}$. Include error bars for $\frac{1}{s}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]

